

ICU Delirium Detection Model

Two-Model Ensemble Approach: Research Summary

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1. Research Background

Objective: Develop a machine learning model to detect current delirium status in ICU patients using an 8-hour observation window.

Key Concept: Since CAM-ICU is recorded periodically by nurses, a positive result indicates that delirium likely existed during the preceding hours. This is a detection approach rather than future prediction.

Data Availability:

	MIMIC-IV	Hospital Data
CAM-ICU Labels	Abundant	Limited
Structured Data	Available	Available
Nursing Notes	Not Available	Available

2. Core Problem: Modality Mismatch

Why use MIMIC? Hospital CAM-ICU labeled data is insufficient for training. MIMIC provides abundant labels for initial model training.

The Dilemma:

- Nursing notes contain valuable information for delirium detection (behavioral changes, disorientation, etc.)
- However, MIMIC lacks nursing notes, making it impossible to train text features
- Features not learned in MIMIC cannot be suddenly used in hospital validation

3. Proposed Approach: Two-Model Ensemble

The key idea is to separate the two models with distinct roles:

Model A: Structured Model (Transferable)

Training	MIMIC-IV (structured features only)
Validation	Hospital (structured features only)
Purpose	Verify transferability from public to local data
Output	P_structured (delirium probability)

Model B: Text Model (Local)

Training	Hospital (nursing notes embedding)
Validation	Hospital internal cross-validation
Purpose	Capture additional predictive power from text
Output	P_text (delirium probability)

Final Ensemble:

$$P_{\text{final}} = \alpha \times P_{\text{structured}} + (1-\alpha) \times P_{\text{text}}$$

where α is optimized on the validation set.

4. How This Approach Solves the Problem

Problem	Solution
No nursing notes in MIMIC	Model A uses structured data only
Limited CAM-ICU labels in hospital	Model A learns from abundant MIMIC labels
Want to utilize nursing notes	Model B learns separately within hospital data
Need external validation	Model A transferability can be purely measured

5. Research Story & Expected Contributions

Key Claims for Publication:

- **Transferability Validation:** Demonstrate that a structured model trained on public data (MIMIC) generalizes to a Korean hospital setting
- **Incremental Benefit of Nursing Notes:** Quantify performance improvement (AUROC increase) when adding nursing notes to the structured baseline
- **Practical Deployment Strategy:** Show that hospitals with limited labeled data can achieve good performance by combining externally-trained models with local text adaptation

Ablation Study Design:

Experiment	Description	Meaning
Model A only	Structured features only	Baseline & transferability
Model B only	Text features only	Nursing notes standalone power
Ensemble	A + B combined	Final performance

6. Study Design Details

Observation Window: 8 hours before each CAM-ICU measurement

Feature Aggregation Strategy (Model A):

Strategy	Description	Use Case
Mean	8-hour average	Overall state
Last	Most recent value	Current state
Min/Max	Extreme values	Abnormal detection
Trend	Start-end difference	State change direction

Text Processing Pipeline (Model B):

Nursing Notes → Small LLM Summary (<20B) → Embedding Model → 150D Vector → Classifier

Evaluation Metrics:

- AUROC (Area Under ROC Curve)
- AUPRC (Area Under Precision-Recall Curve)
- Sensitivity/Recall
- Specificity
- Precision/PPV
- F1 Score

7. Development Phases

Phase	Name	Description	Status
Phase 1	Data Extraction	MIMIC-IV cohort definition, variable extraction	Completed
Phase 2	Preprocessing	8h window, labeling, aggregation, normalization	In Progress
Phase 3	Model A	RF/XGBoost training, MIMIC to hospital validation	Planned
Phase 4	Model B	Nursing notes pipeline, embedding, classifier	Planned
Phase 5	Ensemble	Combine A+B, optimize alpha, ablation study	Planned

8. Key References

1. Bhattacharyya A, et al. (2022). Delirium prediction in the ICU: designing a screening tool for preventive interventions. JAMIA Open, 5(2), ooac048.
2. Ruppert MM, et al. (2020). ICU delirium-prediction models: a systematic review. Crit Care Explor, 2(12), e0296.
3. Chen WH, et al. (2026). Risk prediction models for delirium in ICU patients: a systematic review and critical appraisal. BMC Anesthesiology, 26:43.